

SORTING

PROBLEM 8

QUESTION:

DS	Session	Sorting	Question Information	Level 1	Challenge 18
Question description kkalaiselvan is going to behave as a vehicle driver. he needs to drive an automobile on a track divided into "N" no. of sub-tracks. The letter "K" is also assigned to you. i.e. the maximum number of kilometres a vehicle may go on each sub-track. You can add any unit of Petrol to the automobile if it can't cover a sub-track. The total kilometres your automobile may go increases by one unit for every unit of gasoline added. Input: The first line of input contains an integer T denoting the no of test cases. Then T test cases follow. Each test case contains two space separated integers N and K. The second line of each test case contains N space separated integers (A[]) denoting the distance of each N sub-tracks. Output: For each test case in a new line you have to print out the minimum unit of Petrol your car require to cover all the sub-tracks. If no extra unit of petrol is required, print -1.					

Logical Test Cases

Test Case 1	Test Case 2
INPUT (STDIN) 2 5 6 2 5 4 5 2 5 3 1 6 3 5 2 EXPECTED OUTPUT -1 3	INPUT (STDIN) 2 5 6 12 15 14 15 12 5 3 11 61 31 51 21 EXPECTED OUTPUT 9 58

Mandatory Test Cases

Test Case 1	Test Case 2
KEYWORD void sort(int a[],int n)	KEYWORD for(i=0;i<n-1;i++)

Complexity Test Cases

Test Case 1	Test Case 2	Test Case 3
CYCLOMATIC COMPLEXITY 5	TOKEN COUNT 275	NLOC 40

SOURCE CODE:

```
#include <stdio.h>

int main() {
    int cases;
    if (scanf("%d", &cases) != 1) return 0;

    while (cases--) {
        int segments, fuel_range;
        scanf("%d %d", &segments, &fuel_range);

        int peak_segment = 0;
        for (int i = 0; i < segments; i++) {
            int dist;
            scanf("%d", &dist);
            if (dist > peak_segment) peak_segment = dist;
        }

        if (peak_segment <= fuel_range) printf("-1\n");
        else printf("%d\n", peak_segment - fuel_range);
    }
    return 0;
}
```

OUTPUT:

```
1
5 6
2 5 4 5 2
-1
```

```
2
5 6
12 15 14 15 12
5 3
11 61 31 51 21
9
58
```